# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The perpendicular bisector of a line segment with end points 1,2, 6 and -3,6, 2 passes through -6,2, 4 and has the equation of the form $\frac{x+6}{l} = \frac{y-2}{m} = \frac{z-4}{n}$ (Where $l, m, n$ are integers, $l$ is a prime number and $l > 0$), then the value of $lmn - l + m + n$ equals to

**(A)** $-3$

**(B)** $-5$

**(C)** $-7$

**(D)** $-9$

**Q.2**

What is the perpendicular distance of the point $P(6, 7, 8)$ from $xy$ -plane?

**(A)** 8

**(B)** 7

**(C)** 6

**(D)** None of these

**Q.3**

$L$ is the foot of the perpendicular drawn from a point $P(6, 7, 8)$ on the $xy$ -plane. What are the coordinates of point $L$ ?

**(A)** $(6, 0, 0)$

**(B)** $(6, 7, 0)$

**(C)** $(6, 0, 8)$

**(D)** none of these

**Q.4**

If the origin is shifted $(1, 2 - 3)$ without changing the directions of the axis, then find the new coordinates of the point $(0, 4, 5)$ with respect to new frame.

**(A)** $(-1, 2, 8)$

**(B)** $(4, 5, 1)$

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**(C)** $(3, -2, 4)$

**(D)** $(6, 0, 8)$

**Q.5**

The line, $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the curve $xy = c^2, z = 0$ if $c$ is equal to

**(A)** $\pm 1$

**(B)** $\pm \frac{1}{3}$

**(C)** $\pm \sqrt{5}$

**(D)** None of these

**Q.6**

If two lines $L_1$ and $L_2$ in space, are defined by $L_1 = \{x = \sqrt{\lambda}y + (\sqrt{\lambda} - 1), z = (\sqrt{\lambda} - 1)y + \sqrt{\lambda}\}$ and
$L_2 = \{x = \sqrt{\mu}y + (1 - \sqrt{\mu}), z = (1 - \sqrt{\mu})y + \sqrt{\mu}\}$
then $L_1$ is perpendicular to $L_2$, for all non-negative reals $\lambda$ and $\mu$, such that :

**(A)** $\sqrt{\lambda} + \sqrt{\mu} = 1$

**(B)** $\lambda \neq \mu$

**(C)** $\lambda + \mu = 0$

**(D)** $\lambda = \mu$

**Q.7**

The foot of the perpendicular from $(2, 4, -1)$ to the line
$x + 5 = \frac{1}{4}(y + 3) = -\frac{1}{9}(z - 6)$

**(A)** $(-4, 1, -3)$

**(B)** $(4, -1, -3)$

**(C)** $(-4, -1, 3)$

**(D)** $(-4, -1, -3)$

**Q.8**

The equation of two lines through the origin, which intersect the line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ at angles of $\frac{\pi}{3}$ each, are

**(A)** $\frac{x}{1} = \frac{y}{2} = \frac{z}{1}; \frac{x}{1} = \frac{y}{1} = \frac{z}{2}$

**(B)** $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}; \frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$

**(C)** $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}; \frac{x}{1} = \frac{y}{-1} = \frac{z}{-2}$

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**(D)** None of the above

**Q.9**

The foot of the perpendicular from the point $(1,6,3)$ to the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ is

**(A)** $(1,3,5)$

**(B)** $(-1,-1,-1)$

**(C)** $(2,5,8)$

**(D)** $(-2,-3,-4)$

**Q.10**

From a point $P(\lambda,\lambda,\lambda)$, perpendiculars $PQ$ and $PR$ are drawn, respectively, on the lines $y=x, z=1$ and $y=-x, z=-1$. If $\angle QPR$ is a right angle, then the possible value(s) of $\lambda$ is/are

**(A)** 2

**(B)** 1

**(C)** $-1$

**(D)** $-\sqrt{2}$

**Q.11**

$L_1$ and $L_2$ are two lines whose vector equations are $L_1: \vec{r}=\lambda((\cos\theta+\sqrt{3})\hat{i}+(\sqrt{2}\sin\theta)\hat{j}+(\cos\theta-\sqrt{3})\hat{k}) L_2: \vec{r}=\mu(a\hat{i}+b\hat{j}+c\hat{k})$, where
$\lambda$ and $\mu$ are scalars and $\alpha$ is the acute angle between $L_1$ and $L_2$. If the angle ' $\alpha$ ' is independent of $\theta$ then the value of ' $\alpha$ ' is

**(A)** $\frac{\pi}{6}$

**(B)** $\frac{\pi}{4}$

**(C)** $\frac{\pi}{3}$

**(D)** $\frac{\pi}{2}$

**Q.12**

The line which passes through the origin and intersect the two lines $\frac{x-1}{2}=\frac{y+3}{4}=\frac{z-5}{3}, \frac{x-4}{2}=\frac{y+3}{3}=\frac{z-14}{4}$, is

**(A)** $\frac{x}{1}=\frac{y}{-3}=\frac{z}{5}$

**(B)** $\frac{x}{-1}=\frac{y}{3}=\frac{z}{5}$

**(C)** $\frac{x}{1}=\frac{y}{3}=\frac{z}{-5}$

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**(D)** $\frac{x}{1} = \frac{y}{4} = \frac{z}{-5}$

**Q.13**

Let $L$ be the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$. If $L$ makes an angle $\alpha$ with the positive $x$ -axis, then $\cos\alpha$ equals

**(A)** 1

**(B)** $\frac{1}{\sqrt{2}}$

**(C)** $\frac{1}{\sqrt{3}}$

**(D)** $\frac{1}{2}$

**Q.14**

Under what condition are the two lines
$\mathrm{y} = \frac{\mathrm{m}}{\ell}\mathrm{x} + \alpha, \mathrm{z} = \frac{\mathrm{n}}{\ell}\mathrm{x} + \beta$; and
$\mathrm{y} = \frac{\mathrm{m}'}{\ell'}\mathrm{x} + \alpha', \mathrm{z} = \frac{\mathrm{n}'}{\ell'}\mathrm{x} + \beta'$ orthogonal ?

**(A)** $\alpha\alpha' + \beta\beta' + 1 = 0$

**(B)** $(\alpha+\alpha)+(\beta+\beta)=0$

**(C)** $\ell\ell' + mm' + nn' = 1$

**(D)** $\ell\ell' + mm' + nn' = 0$

**Q.15**

If $\vec{r} = (\hat{i}+2\hat{j}+3\hat{k}) + \lambda(\hat{i}-\hat{j}+\hat{k})$ and
$\vec{r} = (\hat{i}+2\hat{j}+3\hat{k}) + \mu(\hat{i}+\hat{j}-\hat{k})$ are two lines, then
the equation of acute angle bisector of two lines is

**(A)** $\vec{r} = (\hat{i}+2\hat{j}+3\hat{k}) + t(\hat{j}-\hat{k})$

**(B)** $\vec{r} = (\hat{i}+2\hat{j}+3\hat{k}) + t(2\hat{i})$

**(C)** $\vec{r} = (\hat{i}+2\hat{j}+3\hat{k}) + t(\hat{j}+\hat{k})$

**(D)** None of these

**Q.16**

The equation of the line which passes through
the point $(1,1,1)$ and intersect the lines
$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x+2}{1} = \frac{y-3}{2} = \frac{z+1}{4}$ is

**(A)** $\frac{x-1}{3} = \frac{y-1}{10} = \frac{z-1}{17}$

**(B)** $\frac{x-1}{2} = \frac{y-1}{3} = \frac{z-1}{-5}$

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**(C)** $\frac{x-1}{-2}=\frac{y-1}{1}=\frac{z-1}{-4}$

**(D)** $\frac{x-1}{8}=\frac{y-1}{-2}=\frac{z-1}{3}$

**Q.17**

The shortest distance between the skew lines $l_1:\vec{r}=\vec{a}_1+\lambda\overrightarrow{b}_1 l_2:\vec{r}=\vec{a}_2+\mu\overrightarrow{b}_2$ is

**(A)** $\frac{\left|(\vec{a}_2-\vec{a}_1)\cdot\vec{b}_1\times\vec{b}_2\right|}{\left|\vec{b}_1\times\vec{b}_2\right|}$

**(B)** $\frac{\left|(\vec{a}_2-\vec{a}_1)\cdot\vec{a}_2\times\vec{b}_2\right|}{\left|\vec{b}_1\times\vec{b}_2\right|}$

**(C)** $\frac{\left|\left(\vec{a}_2-\vec{b}_2\right)\cdot\vec{a}_1\times\vec{b}_1\right|}{\left|\vec{b}_1\times\vec{b}_2\right|}$

**(D)** $\frac{\left|\left(\vec{a}_1-\vec{b}_2\right)\cdot\vec{b}_1\times\vec{a}_2\right|}{\left|\vec{b}_1\times\vec{a}_2\right|}$

**Q.18**

Let $A(\vec{a})$ and $B(\vec{b})$ be points on two skew line $\vec{r}=\vec{a}+\vec{\lambda}$ and $\vec{r}=\vec{b}+u\vec{q}$ and the shortest distance between the skew line is 1, where $\vec{p}$ and $\vec{q}$ are unit vectors forming adjacent sides of parallelogram enclosing an area of $\frac{1}{2}$ units. If an angle between $AB$ and the line of shortest distance is $60^\circ$, then $AB=$

**(A)** $\frac{1}{2}$

**(B)** 2

**(C)** 1

**(D)** $\lambda\in R-\{0\}$

**Q.19**

Value of $\lambda$ such that the line $\frac{x-1}{2}=\frac{y-1}{3}=\frac{z-1}{\lambda}$ is perpendicular to normal to the plane $\vec{r}\cdot(2\vec{i}+3\vec{j}+4\vec{k})=0$ is

**(A)** $-\frac{13}{4}$

**(B)** $-\frac{17}{4}$

**(C)** 4

**(D)** None of these

**Q.20**

The angle between the straight lines

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$\vec{r} = (2 - 3t)\vec{i} + (1 + 2t)\vec{j} + (2 + 6t)\vec{k}$ and
$\vec{r} = (1 + 4s)\vec{i} + (2 - s)\vec{j} + (8s - 1)\vec{k}$ is

**(A)** $\cos^{-1}\left(\frac{\sqrt{41}}{34}\right)$

**(B)** $\cos^{-1}\left(\frac{21}{34}\right)$

**(C)** $\cos^{-1}\left(\frac{43}{63}\right)$

**(D)** $\cos^{-1}\left(\frac{34}{63}\right)$

## Answers & Solutions

**Q.1 Answer:**

-7

**Solution:**

**Midpoint of the line segment is** $\frac{1-3}{1}, \frac{2+6}{2}, \frac{6+2}{2} \equiv -1, 4, 4$

**Parallel vector to the required line** $= -1 + 6\hat{i} + 4 - 2\hat{j} + 4 - 4\hat{k}$

$= 5\hat{i} + 2\hat{j} + 0\hat{k}$

**Hence, equation of the line is**

$\frac{x+6}{5} - \frac{y-2}{2} = \frac{z-4}{0}$

$\Rightarrow l = 5, \quad m = 2, \quad n = 0$

**Q.2 Answer:**

8

**Solution:**

**Let $L$ be the foot of perpendicular drawn from the point $P(6, 7, 8)$ to the $xy$ -plane and the distance of this foot $L$ from $P$ is $z$ -coordinate of $P$, i.e., 8 units.**

**Q.3 Answer:**

$(6, 7, 0)$

**Solution:**

**Since $L$ is the foot of perpendicular from $P$ on the $xy$ -plane, $z$ -coordinate is zero in the $xy$ -plane. Hence, coordinates of $L$ are $(6, 7, 0)$.**

**Q.4 Answer:**

$(-1, 2, 8)$

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**Solution:**

In the new frame $x' = x - x_1, y' = y - y_1, z' = z - z_1$, where $(x_1, y_1, z_1)$ is shifted origin.
$\Rightarrow x' = 0 - 1 = -1, y' = 4 - 2 = 2 = 2, z' = 5 + 3 = 8$
Hence, the coordinates of the point with respect to the new coordinates frame are $(-1, 2, 8)$

**Q.5 Answer:**

$\pm\sqrt{5}$

**Solution:**

We have, $z = 0$ for the point where the line intersects the curve.
Therefore, $\frac{x-2}{3} = \frac{y+1}{2} = \frac{0-1}{-1}$
$\Rightarrow \frac{x-2}{3} = 1$ and $\frac{y+1}{2} = 1$
$\Rightarrow x = 5$ and $y = 1$
Put these value in $xy = c^2$, we get, $5 = c^2 \Rightarrow c = \pm\sqrt{5}$

**Q.6 Answer:**

$\lambda = \mu$

**Solution:**

For $\mathrm{L}_1$, $x = \sqrt{\lambda}y + (\sqrt{\lambda} - 1) \Rightarrow y = \frac{x-(\sqrt{\lambda}-1)}{\sqrt{\lambda}}$
$z = (\sqrt{\lambda} - 1)y + \sqrt{\lambda} \Rightarrow y = \frac{z-\sqrt{\lambda}}{\sqrt{\lambda}-1}$
From (i) and (ii) $\frac{x-(\sqrt{\lambda}-1)}{\sqrt{\lambda}} = \frac{y-0}{1} = \frac{z-\sqrt{\lambda}}{\sqrt{\lambda}-1}$
The equation (iii) is the equation of line $\mathrm{L}_1$. Similarly equation of line $\mathrm{L}_2$ is $\frac{x-(1-\sqrt{\mu})}{\sqrt{\mu}} = \frac{y-0}{1} = \frac{z-\sqrt{\mu}}{1-\sqrt{\mu}}$
Since $\mathrm{L}_1 \perp \mathrm{L}_2$, therefore $\sqrt{\lambda}\sqrt{\mu} + 1 \times 1 + (\sqrt{\lambda} - 1)(1 - \sqrt{\mu}) = 0$
$\Rightarrow \sqrt{\lambda} + \sqrt{\mu} = 0 \Rightarrow \sqrt{\lambda} = -\sqrt{\mu} \Rightarrow \lambda = \mu$

**Q.7 Answer:**

$(-4, 1, -3)$

**Solution:**

Given equation of line is $x + 5 = \frac{1}{4}(y + 3) = -\frac{1}{9}(z - 6)$
or $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9} = \lambda$( say )
$x = \lambda - 5, y = 4\lambda - 3, z = -9\lambda + 6$
$(x, y, z) \equiv (\lambda - 5, 4\lambda - 3, -9\lambda + 6)$
Let it is foot of perpendicualr So, d.r.'s of $\perp$ line is $(\lambda - 5 - 2, 4\lambda - 3 - 4, -9\lambda + 6 + 1)$
$\equiv (\lambda - 7, 4\lambda - 7, -9\lambda + 7)$
D.r.'s of given line is $(1, 4, -9)$ and both lines are $\perp \therefore (\lambda - 7).1 + (4\lambda - 7).4 + (-9\lambda + 7)(-9) = 0$

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$\Rightarrow 98\lambda = 98 \Rightarrow \lambda = 1$

$\therefore$ **Point is** $(-4, 1, -3)$. $\quad$ [ **Substituting** $\lambda = 1$ **in (i)** ]

**Q.8 Answer:**

$\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}; \frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$

**Solution:**

**Given equation of line is** $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$

$\Rightarrow$ **DR's of the given line are** $2, 1, 1 \Rightarrow$ **DC's of the given line are** $\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}$

**Since, required lines make an angle** $\frac{\pi}{3}$ **with the given line The DC's of the required lines are** $\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}$ **and** $\frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}$ **respectively.**

**Also, both the required lines pass through the origin.** $\therefore$ **Equation of required lines are** $\frac{x}{1} = \frac{y}{2} = \frac{z}{-1}$ **and** $\frac{x}{-1} = \frac{y}{1} = \frac{z}{-2}$

**Q.9 Answer:**

$(1, 3, 5)$

**Solution:**

**Equation of line is** $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ **Any point on this line is** $(K, 2K+1, 3K+2)$**. If this is the foot of perpendicular from** $(1, 6, 3)$ **then** $d.r$ **of this perpendicular are** $< K-1, 2K-5, 3K-1 >$

**Now, using Condition of perpendicularity we have** $(K-1)1 + (2K-5)2 + (3K-1)3 = 0$

$\Rightarrow K - 1 + 4K - 10 + 9K - 3 = 0 \Rightarrow K = 1$

**Hence, Required foot of perpendicular is** $(1, 3, 5)$

**Q.10 Answer:**

$-1$

**Solution:**

**Line 1:** $\frac{x}{1} = \frac{y}{1} = \frac{z-1}{0} = r, Q(r, r, 1)$

**Line 2 :** $\frac{x}{1} = \frac{y}{-1} = \frac{z+1}{0} = k, R(k, -k, -1)$

$\overrightarrow{PQ} = (\lambda - r)\hat{i} + (\lambda - r)\hat{j} + (\lambda - 1)\hat{k}$

$\overrightarrow{PQ}$ **is perpendicular to Line** $1$.

$\Rightarrow \quad \lambda - r + \lambda - r = 0 \Rightarrow \lambda = r$

$\overrightarrow{PR} = (\lambda - k)\hat{i} + (\lambda + k)\hat{j} + (\lambda + 1)\hat{k}$

$\overrightarrow{PR}$ **is perpendicular to Line 2.** $\Rightarrow \quad \lambda - k - \lambda - k = 0 \Rightarrow k = 0$

**Now,** $\overrightarrow{PQ} \perp \overrightarrow{PR}$

$\Rightarrow \quad (\lambda - r)(\lambda - k) + (\lambda - r)(\lambda + k) + (\lambda - 1)(\lambda + 1)$

$= 0$

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$\Rightarrow \quad \lambda = \pm 1$

**For $\lambda = 1$, points $P$ and $Q$ coincide.** $\quad \therefore \lambda = -1$

**Q.11 Answer:**

$\frac{\pi}{6}$

**Solution:**

**Both the lines pass through origin. Line $L$. is parallel to the vector $\overrightarrow{V_1} = (\cos\theta + \sqrt{3})\hat{i} + (\sqrt{2}\sin\theta)\hat{j} + (\cos\theta - \sqrt{3})\hat{k}$**

**and $L_2$ is parallel to the vector**

$\overrightarrow{V_2} = a\hat{i} + \mathrm{b}\hat{j} + \mathrm{c}\hat{k} \quad \therefore \quad \cos\alpha = \frac{\overrightarrow{V_1}\cdot\overrightarrow{V_2}}{\left|\overrightarrow{V_1}\right|\left|\overrightarrow{V_2}\right|}$

$= \frac{a(\cos\theta+\sqrt{3})+(b\sqrt{2})\sin\theta+c(\cos\theta-\sqrt{3})}{\sqrt{a^2+b^2+c^2}\sqrt{(\cos\theta+\sqrt{3})^2+2\sin^2\theta+(\cos\theta-\sqrt{3})^2}}$

$= \frac{(a+c)\cos\theta+b\sqrt{3})+\sin\theta+(a-c-\sqrt{3})}{\sqrt{a^2+b^2+c^2}\sqrt{2+6}}$

**In order that $\cos\alpha$ is independent of $\theta$, we get $a + c = 0$ and $b = 0$**

$\therefore \quad \cos\alpha = \frac{2a\sqrt{3}}{a\sqrt{22}\sqrt{2}} = \frac{\sqrt{3}}{2} \Rightarrow \alpha = \frac{\pi}{6}$

**Q.12 Answer:**

$\frac{x}{1} = \frac{y}{-3} = \frac{z}{5}$

**Solution:**

**Let the line be $\frac{\mathrm{x}}{\mathrm{a}} = \frac{\mathrm{y}}{\mathrm{b}} = \frac{\mathrm{z}}{\mathrm{c}}$**

**If line (i) intersects with the line $\frac{\mathrm{x}-1}{2}$**

**$= \frac{y+3}{4} = \frac{z-5}{3}$, then**

$$\begin{vmatrix} a & b & c \\ 2 & 4 & 3 \\ 4 & -3 & 14 \end{vmatrix} = 0 \Rightarrow 9a - 7b - 10c = 0 \quad \text{...(ii)}$$

**From (i) and (ii), we have $\frac{\mathrm{a}}{1} = \frac{\mathrm{b}}{-3} = \frac{\mathrm{c}}{5}$**

$\therefore \quad$ **The line is $\frac{x}{1} = \frac{y}{-3} = \frac{z}{5}$**

**Q.13 Answer:**

$\frac{1}{\sqrt{3}}$

**Solution:**

**Let the direction cosines of line $L$ be $l, m, n$, then $2l + 3m + n = 0$ ...(i)**

**and $l + 3m + 2n = 0$ ...(ii)**

**on solving equations (i) and (ii), we get**

$\frac{l}{6-3} = \frac{m}{1-4} = \frac{n}{6-3} \Rightarrow \frac{l}{3} = \frac{m}{-3} = \frac{n}{3}$

**Now** $\frac{l}{3} = \frac{m}{-3} = \frac{n}{3} = \frac{\sqrt{l^2+m^2+n^2}}{\sqrt{3^2+(-3)^2+3^2}}$

$\because l^2 + m^2 + n^2 = 1 \quad \therefore \frac{l}{3} = \frac{m}{-3} = \frac{n}{3} = \frac{1}{\sqrt{27}}$

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$\Rightarrow \quad l=\frac{3}{\sqrt{27}}=\frac{1}{\sqrt{3}}, m=-\frac{1}{\sqrt{3}}, n=\frac{1}{\sqrt{3}}$

**Line $L$, makes an angle $\alpha$ with $+$ ve $x$ -axis**

$\therefore l=\cos \alpha \Rightarrow \cos \alpha=\frac{1}{\sqrt{3}}$

**Q.14 Answer:**

$\ell \ell^{\prime}+m m^{\prime}+n n^{\prime}=0$

**Solution:**

**Given two lines are :**

$\mathrm{y}=\frac{\mathrm{mx}}{\ell}+\alpha, \mathrm{z}=\frac{\mathrm{n}}{\ell} \mathrm{x}+\beta$ **and**

$\mathrm{y}=\frac{\mathrm{m}^{\prime}}{\ell^{\prime}} \mathrm{x}+\alpha^{\prime}, \mathrm{z}=\frac{\mathrm{n}^{\prime}}{\ell^{\prime}} \mathrm{x}+\beta^{\prime}$

**These two lines can be represented as :**

$\frac{\mathrm{y}-\alpha}{\mathrm{m} / \ell}=\frac{\mathrm{x}-0}{1}=\frac{\mathrm{z}-\beta}{\mathrm{n} / \ell}$

**and** $\frac{\mathrm{y}-\alpha^{\prime}}{\mathrm{m}^{\prime} / \mathrm{c}^{\prime}}=\frac{\mathrm{x}-0}{1}=\frac{\mathrm{z}-\beta^{\prime}}{\mathrm{n}^{\prime} / \ell^{\prime}}$

**They are orthogonal, if** $\frac{\mathrm{m}}{\ell} \times \frac{\mathrm{m}^{\prime}}{\ell^{\prime}}+1 \times 1+\frac{\mathrm{n}}{\ell} \frac{\mathrm{n}^{\prime}}{\ell^{\prime}}=-1 \Rightarrow \ell \ell^{\prime}+\mathrm{mm}^{\prime}+\mathrm{nn}^{\prime}=0$

**Q.15 Answer:**

$\vec{r}=(\hat{i}+2 \hat{j}+3 \hat{k})+t(\hat{j}-\hat{k})$

**Solution:**

**Lines are** $\vec{r}=(\hat{i}+2 \hat{j}+3 \hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$

**and** $\vec{r}=(\hat{i}+2 \hat{j}+3 \hat{k})+\mu(\hat{i}+\hat{j}-\hat{k})$

**along vectors** $(\hat{i}-\hat{j}+\hat{k})$ **and** $(\hat{i}+\hat{j}-\hat{k})$

**respectively. Angle between two lines**

$=\cos^{-1}\left(\frac{(1) \times(1)+(-1)(1)+(1)(-1)}{\sqrt{3} \sqrt{3}}\right)$

$=\cos^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

**Which is an obtuse angle.**

**- Vector along acute angle bisector**

$=\lambda\left[\frac{\hat{i}-\hat{j}+\hat{k}}{\sqrt{3}}-\frac{\hat{i}+\hat{j}-\hat{k}}{\sqrt{3}}\right]=\frac{2 \lambda}{\sqrt{3}}(-\hat{j}+\hat{k})$

$\therefore \quad$ **Equation of acute angle bisector** $=(\hat{i}+2 \hat{j}+3 \hat{k})+t(\hat{j}-\hat{k})$

**Q.16 Answer:**

$\frac{x-1}{3}=\frac{y-1}{10}=\frac{z-1}{17}$

**Solution:**

**Any line passing through the point** $(1,1,1)$ **is**

$\frac{x-1}{a}=\frac{y-1}{b}=\frac{z-1}{c}$

**This line intersects the line** $\frac{x-1}{2}=\frac{y-2}{3}$

$=\frac{z-3}{4}$

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**If** $a:b:c \neq 2:3:4$ **and** $\begin{vmatrix} 1-1 & 2-1 & 3-1 \\ a & b & c \\ 2 & 3 & 4 \end{vmatrix} = 0$

$\Rightarrow \quad a - 2b + c = 0$

**Again, line (i) intersects line** $\frac{x-(-2)}{1} = \frac{y-3}{2}$

$= \frac{z-(-1)}{4}$

**If** $a:b:c \neq 2:3:4$ **and** $\begin{vmatrix} -2-1 & 3-1 & -1-1 \\ a & b & c \\ 1 & 2 & 4 \end{vmatrix} = 0$

$\Rightarrow 6a + 5b - 4c = 0$

**From (ii) and (iii) by cross multiplication, we have**

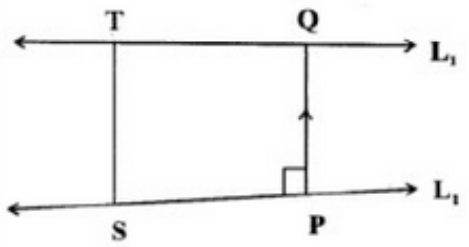
$\frac{a}{8-5} = \frac{b}{6+4} = \frac{c}{5+12}$ **or** $\frac{a}{3} = \frac{b}{10} = \frac{c}{17}$

**So, the required line is** $\frac{x-1}{3} = \frac{y-1}{10} = \frac{z-1}{17}$.

**Q.17 Answer:**

$\frac{\left|(\vec{a}_2 - \vec{a}_1) \cdot \vec{b}_1 \times \vec{b}_2\right|}{\left|\vec{b}_1 \times \vec{b}_2\right|}$

**Solution:**




**(1) Let** $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ **and** $\vec{r} - \vec{a}_2 + \mu\vec{b}_2$ **be two skew lines.**

**(2) Take any point** $S$ **on** $l_1$**, with position vector** $a_1$ **and** $T$ **on** $l_2$ **with the position vector** $a_2$

**(3) clearly vector** $\mathrm{PQ}$ **where**

**where,** $d$**-shortest distance**

$n$ **- units Normal vector**

**(4) Let** $\theta$ **be angle between** $\overrightarrow{\mathrm{PQ}} and \overrightarrow{\mathrm{ST}}$

**Then,** $\mathrm{PQ} = \mathrm{ST}|\cos\theta| - (1)$

**(5) But,**

$$\cos\theta = \left|\frac{\overrightarrow{\mathrm{PQ}} \cdot \overrightarrow{\mathrm{ST}}}{\overrightarrow{\mathrm{PQ}}||\overrightarrow{\mathrm{ST}}|}\right| = \left|\frac{\mathrm{d}\hat{n} \cdot \left(\overrightarrow{\mathrm{a}}_2 - \overrightarrow{\mathrm{a}}_1\right)}{\mathrm{d(ST)}}\right.$$

$$= \left|\frac{\left(\overrightarrow{\mathrm{b}}_1 \times \overrightarrow{\mathrm{b}}_2\right) \cdot \left(\overrightarrow{\mathrm{a}}_2 - \overrightarrow{\mathrm{a}}_1\right)}{\left|\overrightarrow{\mathrm{b}}_1 \times \overrightarrow{\mathrm{b}}_2\right| \mathrm{ST}}\right|$$

**Thus,** $\mathbf{d = PQ} = ST|\cos\theta|$

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$$d = \$T' \left| \frac{\left(\vec{b}_1 \times \vec{b}_2\right) : (\vec{a}_2 - \vec{a}_1)}{\left|\vec{b}_1 \times \vec{b}_2\right| ST'} \right|$$

$$d = \left| \frac{\left(\vec{b}_1 \times \vec{b}_2\right) \cdot (\vec{a}_2 - \vec{a}_1) \mid}{\left|\vec{b}_1 \times \vec{b}_2\right|} \right|$$

**This is the required shortest Distance between two skew lines.**

**Q.18 Answer:**

2

**Solution:**

$1 = \left|(\vec{b} - \vec{a}) \cdot \frac{(\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right| \Rightarrow |\vec{a} - \vec{b}| \cos 60^\circ = 1$

$AB = 2$

**Q.19 Answer:**

$-\frac{13}{4}$

**Solution:**

**Since line is parallel to the plane, vector**

$2\vec{i} + 3\vec{j} + \lambda\vec{k}$ **is perpendicular to the normal to**

**the plane** $2\vec{i} + 3\vec{j} + 4\vec{k}$

$\Rightarrow \quad 2 \times 2 + 3 \times 3 + 4\lambda = 0$ **or** $\lambda = -\frac{13}{4}$

**Q.20 Answer:**

$\cos^{-1}\left(\frac{34}{63}\right)$

**Solution:**

$$\mathrm{L}_1 \equiv \vec{\mathrm{r}}_1 = 2\vec{\mathrm{i}} + \vec{\mathrm{j}} + 2\vec{\mathrm{k}} + \mathrm{t}(-3\vec{\mathrm{i}} + 2\vec{\mathrm{j}} + 6\vec{\mathrm{k}})$$

$$\mathrm{L}_2 \equiv \vec{\mathrm{r}}_2 = (\vec{\mathrm{i}} + 2\vec{\mathrm{j}} - \vec{\mathrm{k}}) + \mathrm{s}(4\vec{\mathrm{i}} - \vec{\mathrm{j}} + 8\vec{\mathrm{k}})$$

**angle between** $\mathrm{L}_1$ **and** $\mathrm{L}_2$ **is given by**

$\cos\theta = \frac{(-3\vec{\mathrm{i}}+2\vec{\mathrm{j}}+6\vec{\mathrm{k}})\cdot(4\vec{\mathrm{i}}-\vec{\mathrm{j}}+8\vec{\mathrm{k}})}{\sqrt{9+4+36}\sqrt{16+1+64}}$

$= \frac{-12-2+48}{\sqrt{49}\sqrt{81}} = \frac{34}{7\times 9} = \frac{34}{63}$

$\Rightarrow \theta = \cos^{-1}\left(\frac{34}{63}\right)$

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